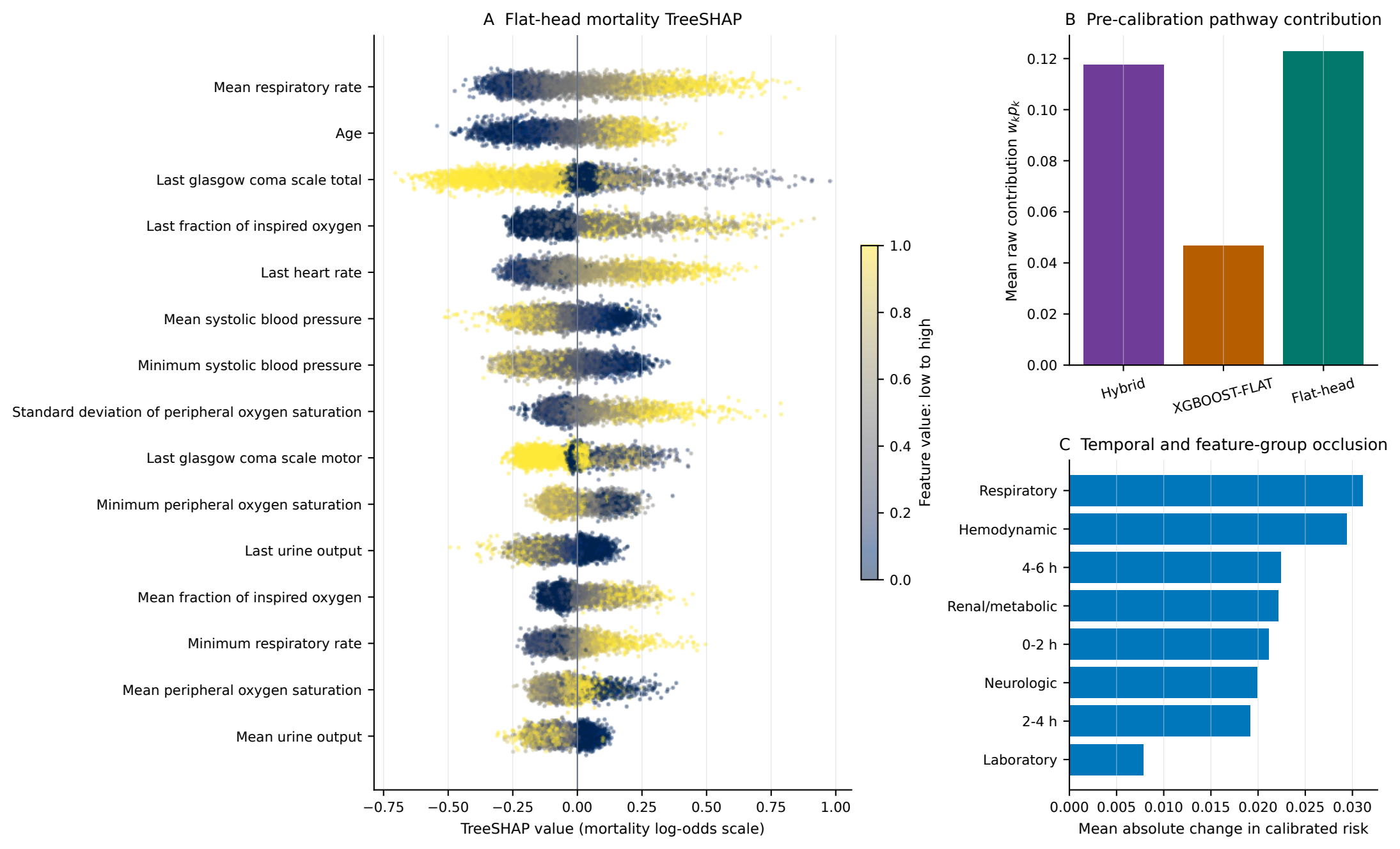

A Flat-head mortality TreeSHAP
Mean respiratory rate
Age
Last glasgow coma scale total
Last fraction of inspired oxygen
Last heart rate
Mean systolic blood pressure
Minimum systolic blood pressure
Standard deviation of peripheral oxygen saturation
Last glasgow coma scale motor
Minimum peripheral oxygen saturation
Last urine output
Mean fraction of inspired oxygen
Minimum respiratory rate
Mean peripheral oxygen saturation
Mean urine output
−0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
TreeSHAP value (mortality log-odds scale)
Feature value: low to high
0.0 0.2 0.4 0.6 0.8 1.0
B Pre-calibration pathway contribution
Mean raw contribution $w_k \rho_k$
0.00 0.02 0.04 0.06 0.08 0.10 0.12
Hybrid
XGBOOST-FLAT
Flat-head
C Temporal and feature-group occlusion
Respiratory
Hemodynamic
4-6 h
Renal/metabolic
0-2 h
Neurologic
2-4 h
Laboratory
0.000 0.005 0.010 0.015 0.020 0.025 0.030
Mean absolute change in calibrated risk